Akandwanaho Ivan

Software engineer & technical founder

Kampala, Uganda | linkedin.com/in/the4codex | github.com/codesphere-u | codesphereug.com

Engineering leadership across real-time marketplaces, mobile-money fraud defense, and AI infrastructure. Hands-on work spans product architecture, application security, and tools that make AI agent behavior reproducible and auditable.

Experience

Founder & Managing Director | Codesphere Innovations (U) Limited (Current)

- Set technology direction and lead product architecture, engineering delivery, and security standards.
- Design web, mobile, and cloud systems across applied AI, business workflows, and custom software.

Founder & Project Lead, Funa Gigs | Jaminsa Ventures Limited (Current)

- Lead engineering for a Ugandan services marketplace available on Android and the web.
- Oversee location-aware Instant Hire, in-app coordination, and payment release after job approval, with MTN and Airtel Money withdrawals.

CTO & Head of Engineering | Waanverse Labs Inc. (Until March 2025)

- Directed technical strategy, architecture, and engineering delivery. Defined API standards, guided code reviews, and established release workflows.

Founder & Lead Software Engineer | 4codex Technologies (Earlier venture)

- Built business applications and automated workflows. Delivered application security hardening and technical consulting.

Selected engineering

Cyber Dragon - Built a workbench that explains signs of mobile-money fraud, USSD traps, and deceptive URLs without fetching submitted links.

AgentFlight - Built deterministic replay for AI tool calls. It reuses recorded responses and reports when inputs diverge from a saved run.

AgentShield - Built a runtime firewall for AI agents and MCP tools with capability policies, outbound data checks, and a tamper-evident audit trail.

FlowLedger - Built a Python engine that models payment retries and out-of-order webhooks while enforcing balanced double-entry journal entries.

StackPulse - Built a trace analysis toolkit for following requests across services and locating failure paths with an instrumented simulator.

Technical skills

Languages & frameworks: Python, TypeScript, Dart, SQL, FastAPI, React, Next.js, Flutter

Data & delivery: PostgreSQL, SQLite, Docker, OpenTelemetry, GitHub Actions

Systems design: API architecture, application security, agent replay, payment reconciliation

Education

ISBAT University | Kampala, Uganda

University studies alongside independent work building software for local businesses.